Open-Source 3D Printed Demonstrations for Visualizing Transverse Wave Motion

Rachel Lee, Abhishek K. Sharma

Department of Chemical Engineering, The Cooper Union for the Advancement of Science and Art, New York, New York 10013

MATERIALS + EXPRESSION × TECHNOLOGY + ART

Introduction

- The use of 3D printed science demonstrations have grown in popularity in K-12 science education due to their usefulness and ease of use.
- 3D printed demonstrations have been shown to significantly decrease instructors' teaching anxiety, improve teaching efficacy, and increase student engagement as new visual aids and means of tactile learning [1, 2]
- The demonstrations consist of a central axis and spokes that oscillate to represent the perpendicular displacement characteristic of transverse waves
- ***Key design parameters:*** **overall size, spoke geometry, axis geometry, and support structure**
- ***Evaluation parameters:*** **printability, mechanical stability, and visual clarity of motion, and cost-effectiveness**

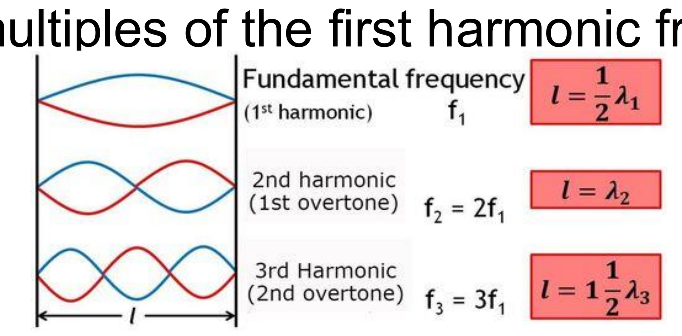
How do the demos work?

- The first harmonic frequency is given by

$$f = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

where L is the length of the spine, T is tension, and μ is the mass per unit length of the string

- Succeeding harmonics have frequencies that are multiples of the first harmonic frequency [3]



Commercially available sound wave demo offered by Pasco Scientific, priced at over $1,000.

Methods

Drawing and Designing

- Designs were created and modeled using OnShape and Autodesk Fusion 360.
- Various axis types were developed, including chain, rod, tape, spring, and multi-axis configurations.
- Spoke designs consisted of both disc and bar geometries. Supports were designed in three forms: solid, frame, and line structures.

Printing

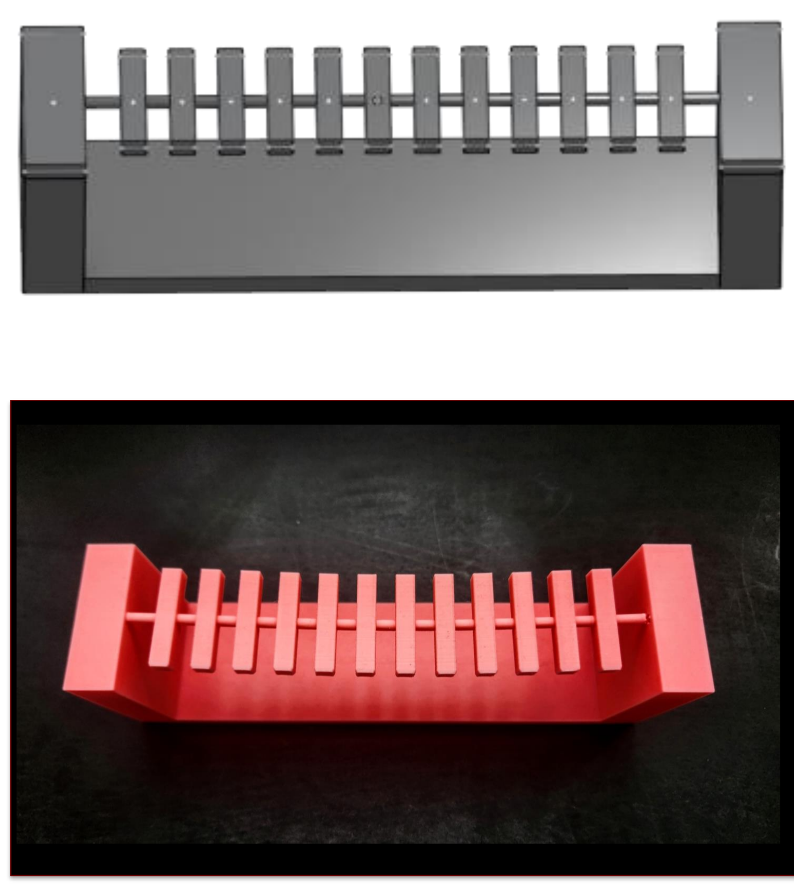
- We used PLA filament with gyroid infill (15%) and a layer height of 0.2 mm.
- Models were printed using the Ultimaker S5, Bambu Lab X1 Carbon Printer, or Prusa MK4S

Ultimaker

Bambu Lab

Results and Discussion

Design 1

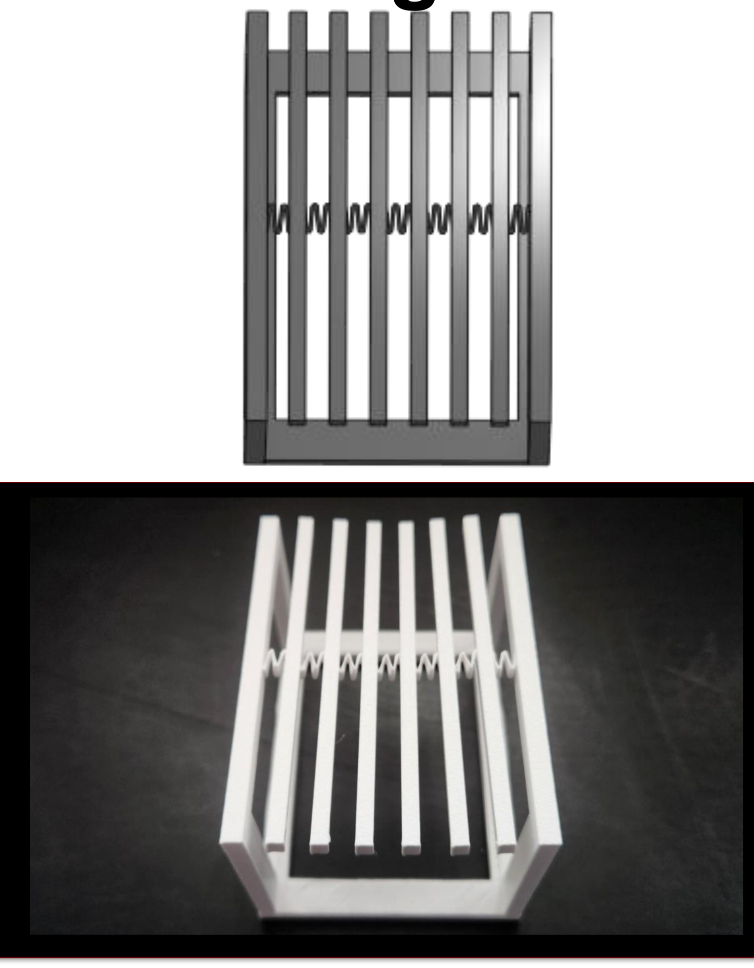


Rod and bars design:
- Rectangular prism bars
- Solid base
- *Pros:* easy and quick to print, highly stable design
- *Cons:* waves difficult to observe due to high frequency and stiffness

Design 2

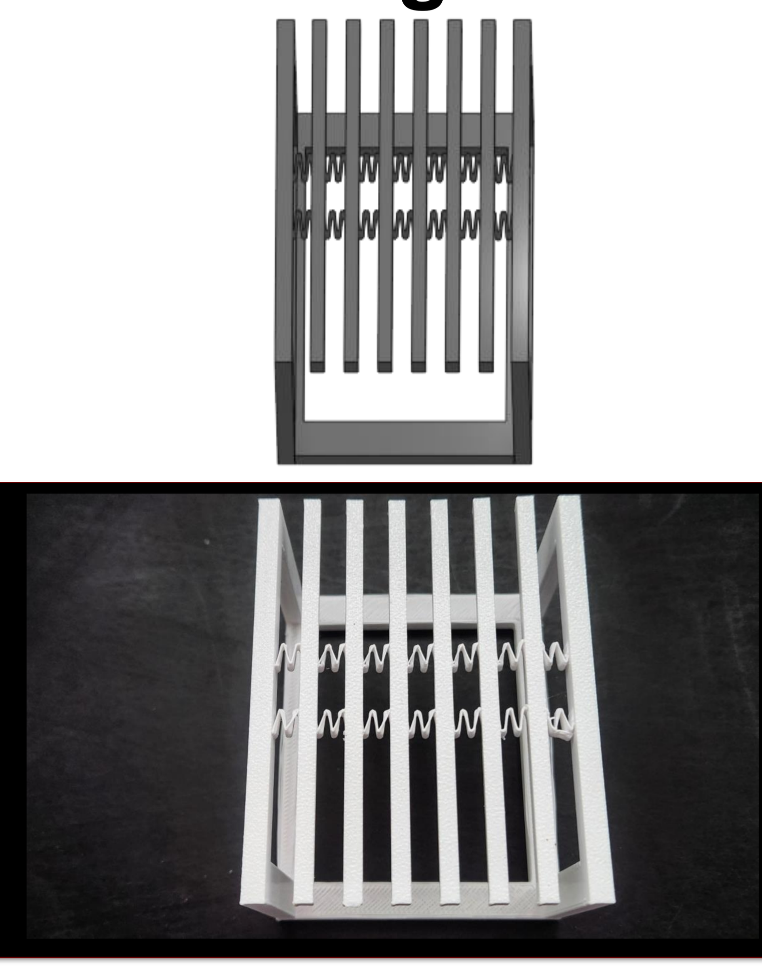
Tape and bars design:
- *Pros:* tape was more flexible and less stiff compared to rod, resulting in waves that were larger in amplitude and lower in frequency; base design was altered to reduce material costs; increased length of bars resulted in greater inertia and torque
- Cons: waves still difficult to observe due to high frequency

Design 3



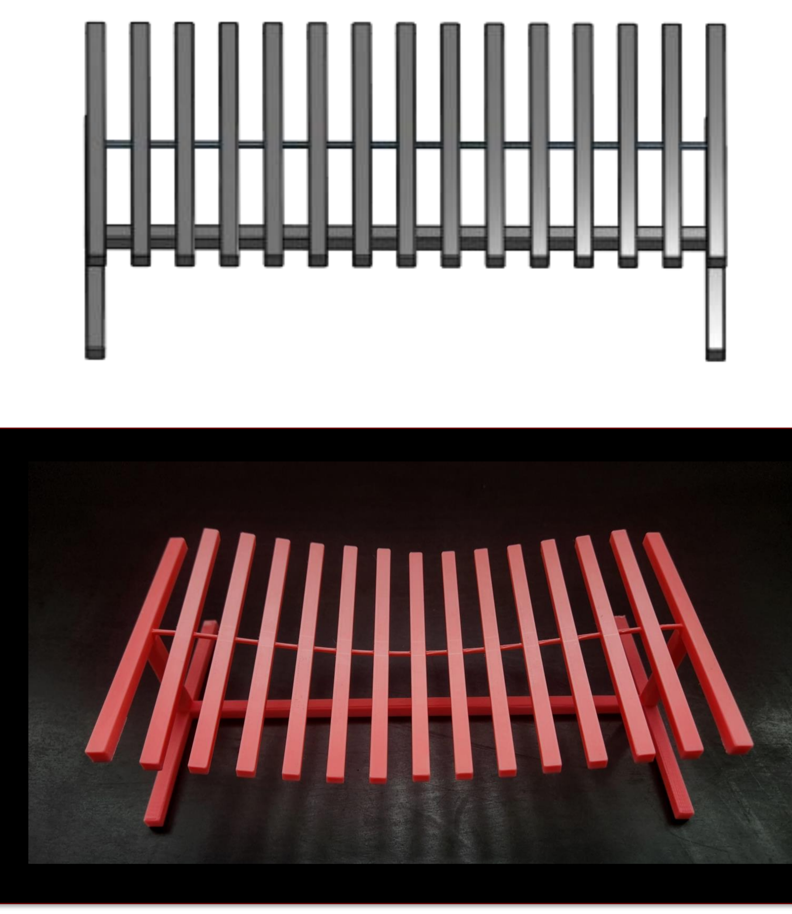
Single horizontal spring design:
- *Pros:* most flexible and sensitive design
- *Cons:* bars exhibited longitudinal motion, rather than transverse motion; springs were delicate and easily broke

Design 4



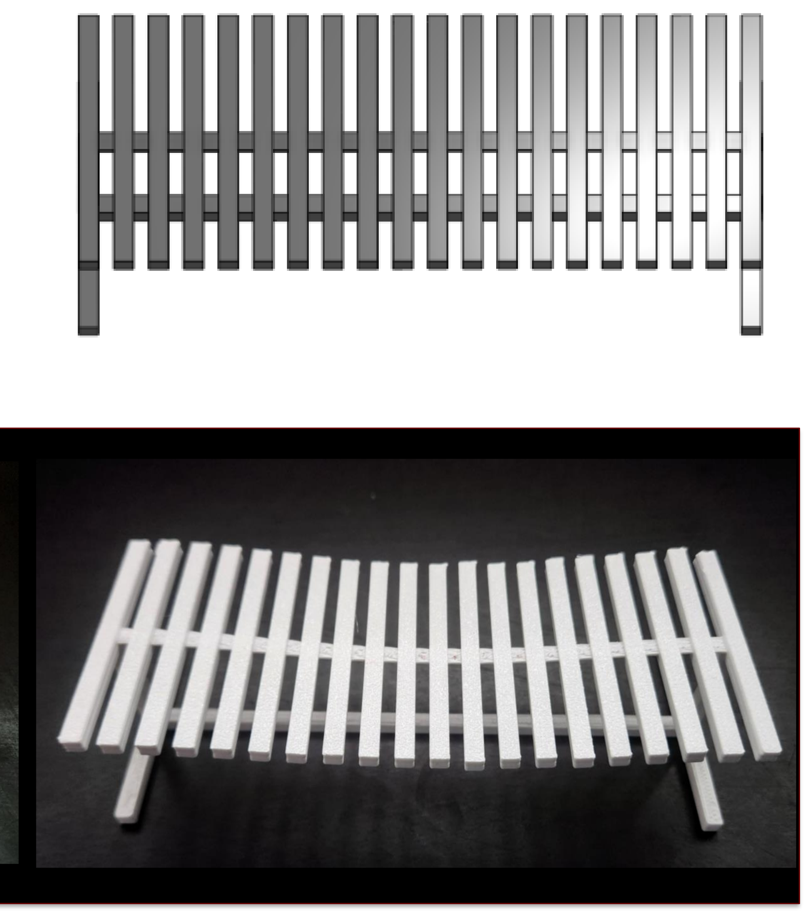
Double horizontal springs design:
- *Pros:* increased stiffness and resilience, compared to single spring design
- *Cons:* bars exhibited longitudinal wave motion

Design 5



Thin rod and bars design:
- *Pros:* base design altered to reduce material costs and printing time; the length of the model was increased, the diameter of the rod decreased, and the length of the bars increased, resulting in waves of larger amplitudes and lower frequencies
- *Cons:* extremely delicate, due to thinness of rod

Design 6



Thin tape and bars design:
- *Pros:* the length of the model was increased and a thin tape was used, resulting in a more resilient design with observable transverse waves
- *Cons:* due to the increased length of the model, tension along the bars was lost

Future Work

We plan to (1) parametrize our designs to make them easily adjustable and adaptable and (2) improve the modularity of our designs to allow for easier assembly and modification. All design files will be made open-source and freely available on platforms such as GitHub.

References

1) Novak, E.; Wisdom, S. Effects of 3D Printing Project-Based Learning on Preservice Elementary Teachers' Science Attitudes, Science Content Knowledge, and Anxiety About Teaching Science. *J Sci Educ Technol* **2018**, *27* (5), 412–432.

2) Pinger, C. W.; Geiger, M. K.; Spence, D. M. Applications of 3D-Printing for Improving Chemistry Education. *J. Chem. Educ.* **2020**, *97* (1), 112–117.

3) Standing Waves on a String. HVY Science. https://hvysci.weebly.com/on-a-string.html.

Acknowledgments

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